

SATWIK BHATNAGAR

GRADUATE FULL-STACK SOFTWARE ENGINEER

Full UK work rights (Graduate visa, no sponsorship required)

London, UK | +44 7352 668970 | sa7wik@gmail.com

[Github](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

London-based Full-Stack Software Engineer specializing in scalable web architecture, secure authentication flows, and modern AI/ML integrations, backed by an MSc in Software Engineering with Management Studies. I design and ship production systems end-to-end, from typed **React** front-ends to **Node/Express** and **PostgreSQL** back-ends deployed through automated **CI/CD pipelines** on **AWS** and **Render**. I combine deep engineering rigor, including a from-scratch **C++ streaming server** and enterprise-grade **JWT** security flows, with a product mindset focused on performance, reliability and clean user experience.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, Java, SQL, HTML/CSS

Front-End: React, Vite, Tailwind CSS, Bootstrap, Three.js, Cornerstone.js, Responsive UI, Core Web Vitals

Back-End & Databases: Node.js, Express, Flask, REST APIs, SQLAlchemy, PostgreSQL (Neon), JWT Authentication

Cloud & DevOps: AWS (EC2, S3), Vercel, Render, GitHub Actions (CI/CD), Caddy, Gunicorn, systemd, Git

AI/ML Integration: Claude, Google Gemini, Groq, Whisper, llama.cpp, scikit-learn, TF-IDF

Tools & Methodologies: Agile/Scrum, CMake, PyInstaller, Inno Setup, MinGW-w64, MSYS2, Adobe Premiere Pro & Photoshop

PROJECTS

MediSync: Enterprise-Grade Hospital Management Platform

medisync-hms.onrender.com

Tech: React, TypeScript, Express, PostgreSQL (Neon), AWS S3, Cornerstone.js, JWT, Tailwind CSS, Vite, GitHub Actions, Render.

- Built a full-stack TypeScript platform spanning **69 REST endpoints** across 15 route files and an **11-table PostgreSQL schema**, serving four staff roles (admin, doctor, receptionist, radiologist) plus a separate patient portal covering appointments, records, vitals, prescriptions and billing.
- Engineered a dual-authentication system isolating staff and patients across separate JWT token types (**httpOnly cookies** vs **Bearer tokens**), hashing credentials with **bcrypt at 12 rounds** and enforcing 8-hour token expiry, per-IP **rate limiting** (5 requests / 15 minutes) and a 5-attempt account lockout.
- Built an in-browser **DICOM** medical-image viewer (**Cornerstone.js**) streaming scans across four modalities (X-Ray, MRI, CT and ultrasound) from a private **AWS S3** bucket, with zoom, pan and window-level controls.
- Shipped to production through a **GitHub Actions CI/CD** pipeline gated by **68 passing tests** (47 backend Jest, 21 frontend Vitest), backed by **15 versioned migrations** on **Neon PostgreSQL** deployed on **Render**.

Zathas: Multimodal AI Chatbot with a From-Scratch C++ Server

zathas.com

Tech: C++, cpp-httplib, React, Vite, Tailwind CSS, Groq, Whisper, AWS EC2, Caddy.

- Engineered a from-scratch **C++ server** (cpp-httplib, **1,754 lines across 16 files**) that compiles to a single binary serving five API endpoints and the React front-end from one process, demonstrating systems-level control over routing, streaming and static hosting.
- Streamed live model responses at a measured **~290 ms first-token latency** (five-run benchmark over the public internet), by implementing **Server-Sent Events (SSE) token streaming** with live markdown parsing and syntax highlighting.
- Enabled multimodal input across **32 file types** (images, audio via **Whisper**, text and scanned PDFs, Office .docx/.xlsx/.pptx), by hand-writing C++ parsers for Office (ZIP/XML) and PDF text extraction and routing vision and audio to **Groq**.
- Deployed on **AWS EC2** with auto-renewing HTTPS (Let's Encrypt via a **Caddy reverse proxy**), systemd auto-restart and per-IP **rate limiting** (25 messages/min, 15 uploads/min, sliding window).

NeuroMetric: AI-Powered Subjective Exam Portal

neurometric-mtxc.onrender.com

Tech: Python, Flask, SQLAlchemy, PostgreSQL, Google Gemini, scikit-learn, Bootstrap, Gunicorn, Render.

- Built a multi-tenant Flask exam platform spanning **30 URL routes** across four blueprints (auth, teacher, student, exam) and a **9-table PostgreSQL schema** modelled with **SQLAlchemy**, with role-based teacher and student access across multiple institutions.
- Automated subjective grading across three question types (short, long, essay) with **Google Gemini 2.5 Flash**, backed by a resilient retry layer of **5 attempts with exponential backoff** (capped at 30s plus jitter) to absorb API failures.
- Engineered an academic-integrity engine tracking **three signals per answer** (tab switches, paste events and an AI-authorship score from 0 to 100), auto-flagging AI-written answers above a score of 70 and cross-student plagiarism at a **TF-IDF cosine-similarity threshold of 0.85**.
- Captured live exam-integrity events from the browser through a real-time **POST /exam/log_event** JSON endpoint, and gave teachers a review dashboard with **11-column CSV export** per submission row.

Smart File Organizer: Local-First, On-Device AI Desktop App

github.com/zatwik511/File-Organizer

Tech: C++, llama.cpp, Phi-3 Mini (GGUF), CMake, Python, Tkinter, Inno Setup.

- Classified files entirely offline at zero API cost, by running a quantized **Phi-3 Mini** model (3.82B params, Q4_K_M, **2.23 GiB**) through **llama.cpp** on CPU, batching **100 files per inference call** into 11 categories spanning 112 file extensions.

- Paired the AI mode with six rule-based GUI sort modes benchmarked at **0.58 µs per file** (~0.6 ms for 1,000 files), and shipped the combined C++/Python app as an **11.8 MB standalone Windows executable** with a one-click **Inno Setup** installer, dry-run preview and JSON-backed undo.

TECHNICAL EXPERIENCE

WordPress Developer, Shad Inc. (January - March 2026)

- Delivered client websites end-to-end, by gathering requirements directly from stakeholders and owning the full component lifecycle from design through to deployment.
- Improved **Core Web Vitals** and page-load speed by optimizing responsive layouts, asset delivery and caching across desktop and mobile.
- Increased organic traffic by implementing on-page **SEO** best practices and resolving cross-browser and mobile rendering defects.
- Built bespoke themes and plugins with PHP, HTML, CSS and JavaScript, integrating Elementor, Advanced Custom Fields and **WooCommerce** for e-commerce stores.

Freelance Full-Stack Web Developer (March - May 2026)

- Shipped full-stack websites from the ground up, by designing both front-end and back-end and managing hosting, deployment and ongoing maintenance.
- Lifted client engagement by translating briefs into clean, responsive UIs optimized for performance and **SEO**.
- Reduced post-launch defects by establishing structured testing and release workflows.

AI Data & Model Evaluation, Mercor (May - June 2026)

Multimodal AI Evaluator

- Validated multimodal model outputs against structured rubrics and improved training-data quality by systematically comparing and annotating audio, image and video generations for a leading AI research lab.
- Maintained labelling consistency, by executing standardized evaluation test suites independently to a high-quality bar.

Gameplay Data Contributor

- Synthesized high-fidelity behavioral datasets for model training, by capturing structured gameplay sessions with synchronized screen and input logging.
- Ensured recorded assets passed QA, by enforcing strict capture configuration (stable FPS, correct settings) and managing secure metadata uploads to cloud storage.

EDUCATION

MSc Software Engineering with Management Studies, Kingston University London (2025–2026)

- **Key modules:** Agile Project Development; Software Architecture and Programming; Software Quality Engineering; Business in Practice.
- **Dissertation, *Lean Pipeline Bottleneck Quantification (LPBQ)*:** a data-driven framework that detects, measures and ranks bottlenecks in Agile delivery pipelines. Ingests workflow history from **Jira, GitHub and Azure DevOps**, applies **process mining (PM4Py)**, statistical process control and Lean flow metrics (cycle time, lead time, throughput, flow efficiency), and surfaces remediations through a **Power BI** dashboard. Directly applicable to modern **DevOps and CI/CD** optimization.

B. Tech Computer Science Engineering, Chandigarh University (2019–2023)

- **Coursework:** Data Structures, DBMS, Networking, Operating Systems, Computer Vision, AI/ML, Microprocessors, Graphics, Blockchain, Mobile App Development.
- Final-year project: Evaluation of Subjective Answers using NLP (later rebuilt as the NeuroMetric platform above).

CERTIFICATIONS

- **Machine Learning Specialization**, University of Washington (Coursera), 2025. Regression, classification, clustering and information retrieval.
- **Generative AI: Introduction and Applications**, IBM (Coursera), 2025. LLMs and generative tools across text, image, code and audio.
- **Foundations of Digital Marketing & E-commerce**, Google (Coursera), 2023.